



# Topology-driven tuning of nonlinear opto-electronic properties in spiral-distorted quantum rings under magnetic field

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## Highlights

- First combined study of spiral dislocation ( $\beta$ ), confinement radius ( $R_{\min}$ ) and magnetic field ( $B$ ) effects in 2D quantum rings.
- Magnetic field and spiral dislocation show antagonistic influence on nonlinear optical responses.
- Structural geometry enhances oscillator strength, while dislocation gradients reduce it.
- Coupled ( $\beta$ - $R_{\min}$ - $B$ ) tuning enables optimization of refractive-index changes for QR switching devices.

## Article preview

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## Abstract

This work unlocks the joint effects of radial spiral dislocation ( $\beta$ ), the radial position of the confinement minimum ( $R_{\min}$ ) and an external applied magnetic field ( $B$ ) on the electron nonlinear optical properties of a charged spinless particle confined in a 2D quantum ring (QR). The modified Schrödinger equation has been solved through a self-consistent technique. Within the framework of the density matrix formalism, the linear and nonlinear optical absorption coefficients as well as refractive index changes are evaluated. Our calculations reveal that the electron wave function, dipole matrix elements, oscillator strength, and optical responses are highly responsive to the tested parameters. In particular, both the magnetic field and spiral dislocation act in an antagonist way, while the radial minimum position seems to mask the torsional deformation consequences when enlarged. The oscillator strength is strongly enhanced by the structural geometry, while a radial dislocation gradient decreases its magnitude. Two-dimensional colormaps have deciphered the complex relationship between the combination ( $\beta$ - $R_{\min}$ - $B$ ) on refractive index changes, suggesting the possibility of nonlinear spectrum optimization for converter switching future QR devices.

## Keywords

Quantum ring; Spiral deformation; SACs; RRIAs and oscillator strength

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## Introduction

For physical systems, topological defects must be investigated by condensed matter and nanoscience physicists [1]. Among the defects that may arise are dislocations and deformations caused by torsion, bending and many other constraints [2]. A structure that undergoes curvature exhibits physicochemical properties different from those of an ideally nonperturbed system. Such defects strongly impact the aspects of electronic wave functions, whose perception of the new potential landscape (quantum barriers and trapping zones) becomes different [3]. On the experimental level, dislocation defects induced by applied constraints lead to consequences that can shift numerical predictions. One phenomenon that is closely related to these structures is electronic transport from one region to another in optoelectronic devices involving semiconductors, junctions, and semiconductor–insulator interfaces [4].

Spiral deformations (torsional defects) constitute a particular class of topological defects characterized by a local torsion of the crystalline lattice without macroscopic curvature of space [5]. Unlike ordinary elastic deformations, that relies on displacement gradients, such specific structures introduce a continuous rotation of local reference frames around a given axis or a topological core. Within the geometric framework of analog gravity, reporters proved that defects are described by a nonzero torsion tensor  $T_{\mu\nu}^{\lambda}$  while the curvature tensor remains zero [6]. This fundamentally distinguishes them from disclinations associated with effective curvature. This non-Riemannian geometry profoundly modifies the effective metric experienced by charge carriers and acts as a source of geometric fields influencing their wave functions [7].

Spiral deformations may emerge in nanostructures subjected to constraints that are incompatible with simple elastic relaxation. In QD-types like core/shell or core/multishell systems [8]. Lattice parameter mismatch between the different layers induces internal strain that, beyond a critical threshold, relaxes through the formation of screw dislocations and helical torsion fields. Similarly, in graphene and two-dimensional materials, thermal gradients, localized doping, charged impurities, or transverse electric fields can generate a local symmetry breaking that leads to a rotation of the crystallographic axes [9]. These mechanisms produce an effective chiral geometry that acts as a pseudo-magnetic field and imposes a long-lasting topological constraint on the system [10]. In a space endowed with torsion, the Schrödinger equation acquires a geometric connection term equivalent to an effective vector potential  $A_{\text{torsion}}$ . This geometric field modifies the phase of the wave function and induces a helical quantization of the eigenstates [11]. As a result, the degeneracy of energy levels is lifted, charge carriers become directionally localized, and geometric phases analogous to the Aharonov–Bohm effect appear. In confined systems, such as quantum dots or cylindrical heterostructures, these effects lead to a

measurable shift of electronic transitions and a direct modification of the density of states [12].

Researchers in recent years have paid much attention to studying the nonlinear optical properties of quantum dots, quantum rings, and quantum wells [13,14]. Spiral (torsional) distortions in quantum rings represent a particular case where geometric and topological effects directly influence the confinement landscape. In such systems, the presence of a screw dislocation leads to a coupling between radial and angular coordinates, resulting in a non-Euclidean geometry. This modification alters the electronic structure, transition energies, and optical response of the system [15]. In particular, torsion-induced effects can significantly impact dipole transitions, oscillator strengths, and nonlinear optical coefficients [16].

From a broader perspective, similar geometry-driven effects have been extensively investigated in nonlinear optical media and structured photonic systems, where symmetry breaking and spatial modulation play a crucial role in enhancing light-matter interactions [17,18]. In particular, the control of nonlinear responses through structural ordering and modulation has been demonstrated in advanced optical platforms [19]. Likewise, engineered photonic systems exhibiting high-quality resonances highlight the importance of geometry in tailoring spectral properties and field localization [20]. In ring-like and toroidal configurations, confinement and boundary conditions further amplify interference effects, leading to enhanced sensitivity of optical responses [21]. Moreover, nonlinear responses in circular structures underline the general impact of ring geometries on wave dynamics and energy transfer processes [22]. As a consequence, spiral distortions provide an additional degree of freedom to engineer and tune nonlinear optical responses. These properties make spiral-distorted quantum rings promising platforms for next-generation optoelectronic and photonic devices, with potential implications for nanoscale sensing, bio-photonic systems, and advanced optical diagnostics [23,24]. Depending on the degrees of freedom, the electronic wave functions are manipulated at the nanoscale to satisfy industrial applications. A spatial or partial overlap is the starting point for acquiring the orders of magnitude of nonlinear attributes [25]. As an immediate consequence of geometric restriction, quantum confinement becomes more significant, and second harmonic and third harmonic generation become more appreciated both from the point of view of spectral position and the optimization of certain wavelengths that are otherwise inaccessible to obtain [26].

Choubani et al. [27] have reported the effects of transverse electric field, pressure, and temperature on the nonlinear optical properties of lens-shaped core/shell quantum dots coupled with a wetting layer. They proved that manipulating geometrical metric could enhance the resonance frequencies. Karim et al. [28] have deciphered the strain effects on the band gap and diameter of CdSe core and CdSe/ZnS core/shell quantum dots at any temperature. They indicated that temperature and dimension should be studied in parallel in order to obtain the desired nonlinear optical properties. Fakkahi et al. [29] reported that temperature and pressure represent effective tools to achieve tunable second harmonic generation in GaAs/AlGaAs multilayered quantum dots. They also demonstrate that enlarging the inner core radius redshifted the computed coefficients with slight incrementation in their amplitudes.

In this innovative work, we aim to clarify and examine the effect of the presence of a radial distortion at the confinement trapping well on the electronic wave functions,

the dipole matrix element, the oscillator strength and the absorption/refraction spectra. The originality of this work lies in the fact that most theoretical studies assume a homogeneous confinement environment, although experimental results show quite the opposite. Through this study, we aim to investigate the parallel effect of spiral distortion, the magnetic field and the minimum of the confinement region. The paper is organized into four main parts: the introduction is presented in Section (I). The part containing the mathematical equations as well as the light-matter observables is grouped in Section (II). The interpretation of the major outcomes are presented in Section (III). Our investigation concludes with Section (IV) where we summarize all the concluding points regarding our research.

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## Section snippets

### Geometrical overview

In ordinary Euclidean space (no defect), cylindrical coordinates  $(r, \theta, z)$  have the metric:

$$ds^2 = dr^2 + r^2 d\theta^2 + dz^2$$

A screw dislocation corresponds to a topological defect in which a rotation around the symmetry axis (generally  $zz'$ ) is accompanied by a longitudinal shift of the medium along the same axis. In this geometry, a complete rotation  $\theta \rightarrow \theta + 2\pi$  produces a translation parallel to  $z$ . As a result, the space acquires a helical structure rather than a purely cylindrical one.

Two cases are to consider.

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### Discussion of results

We begin our analysis by Fig. 3 which is a 3D geometric distortion induced by a spiral dislocation confinement potential in case of distinct  $\beta$  values. The existence of the spiral dislocation introduces corrections thereby altering the effective contributions

from the magnetic and centrifugal terms [41]. When  $\beta$  gets greater, the potential becomes increasingly asymmetric and we clearly distinguish drastic modifications to the curvature near the center. We will see later how does such geometric ...

## Conclusion

In this work, we have investigated the combined effects of radial spiral dislocation, the radial position of the confinement minimum, and an external magnetic field on the nonlinear optical properties of a charged spinless particle confined in a two-dimensional quantum ring. Within the effective mass approximation, the modified Schrödinger equation has been solved using a self-consistent approach, while the optical responses have been evaluated through the density matrix formalism. The results ...

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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